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LESSON PLAN	COURSE: MHF4U
Date: Lesson #10	**Unit of Study: Chapter#5 Trigonometric Functions**
Materials / Resources: - **Advanced Function 12 Study Guide** - **Chapter 4 & 5 Formula, Aid and Homework Sheet(s)** - **KWL Chart** - **To solve for trigonometric equations:** [**www.symbolab.com/solver/trigonometric-equation-calculator**](www.symbolab.com/solver/trigonometric-equation-calculator)	**Lesson Focus** - **Graphs of Reciprocal Trigonometric Functions** - **Sinusoidal Functions of the Form** $f(x) = a\cos cos\,[k(x-d)] \quad + c$ - **Solving Trigonometric Equations**

Prior Knowledge:

- Students are able to describe key properties of periodic functions
- Understandings transformations of a function
- Students are able to solve simple trigonometric equations (e.g. sin $sin\,x \;= \frac{1}{2}$)

Overall Expectations:

1. Make connections between trigonometric ratios and the graphical and algebraic representations of the corresponding trigonometric functions and between trigonometric functions and their reciprocals, and use these connections to solve problems
2. Solve problems involving trigonometric equations and prove trigonometric identities

Specific Expectations:

2.3 Graph, with technology and using the primary trigonometric functions, the reciprocal trigonometric functions (i.e., cosecant, secant, cotangent) for angle measures expressed in radians, determine and describe key properties of the reciprocal functions (e.g., state the domain, range, and period, and identify and explain the occurrence of asymptotes), and recognize notations used to represent the reciprocal functions [e.g., the reciprocal of $f(x) = \sin sin\,x$ can be represented using csc $csc\,x$, $\frac{1}{f(x)}$, or $\frac{1}{\sin sin\,x}$, but not using $f^{-1}(x)$ or x , which represent the inverse function]

2.4 Determine the amplitude, period, and phase shift of sinusoidal functions whose equations are given in the form $f(x) = a\sin sin\,(k(x-d)) \;+ corf(x) = a\cos cos\,(k(x-d)) \;+ c$, with angles expressed in radians

2.5 Sketch graphs of $y = a\sin sin\,(k(x-d)) \;+ c$ and $y = a\cos cos\,(k(x-d)) \;+ c$ by applying transformations to the graphs of $f(x) = \sin sin\,x$ and $f(x) = \cos cos\,x$ with angles expressed in radians, and state the period, amplitude, and phase shift of the transformed function

2.6 Represent a sinusoidal function with an equation, given its graph or its properties, with angles expressed in radians

2.7 Pose problems based on applications involving a trigonometric function with domain expressed in radians (e.g., seasonal changes in temperature, heights of tides, hours of daylight, displacements for oscillating springs), and solve these and other such problems by using a given graph or a graph generated with or without technology from a table of values or from its equation

3.4 Solve linear and quadratic trigonometric equations, with and without graphing technology, for the domain of real values from 0 to 2π, and solve related problems

Learning Goals:

- By the end of this lesson I will be able to determine amplitude, vertical translation, phase shift, and period of sinusoidal function given its amplitude
- By the end of this lesson I will be able to determine the equation of a sinusoidal function given its amplitude, vertical translation, phase shift and period
- By the end of this lesson I will be able to determine angles that have a given sine, cosine, tangent, cosecant, secant, or cotangent
- By the end of this lesson I will be able to solve linear and quadratic trigonometric equations
- By the end of this lesson I will be able to model real-world situations using reciprocal trigonometric equations
- By the end of this lesson I will be able to given a data set, model and the data using a sinusoidal function
- By the end of this lesson I will be able to determine values for inverse trigonometric relations

Success Criteria:

1. By the end of this lesson I will be able to determine amplitude, vertical translation, phase shift, and period of sinusoidal function given its amplitude
2. By the end of this lesson I will be able to determine the equation of a sinusoidal function given its amplitude, vertical translation, phase shift and period
3. By the end of this lesson I will be able to determine angles that have a given sine, cosine, tangent, cosecant, secant, or cotangent
4. By the end of this lesson I will be able to solve linear and quadratic trigonometric equations
5. By the end of this lesson I will be able to model real-world situations using reciprocal trigonometric equations
6. By the end of this lesson I will be able togiven a data set, model and the data using a sinusoidal function
7. By the end of this lesson I will be able to determine values for inverse trigonometric relations

Assessment Strategies:

<u>Formative</u>: homework for chapters #5.2, 5.3 & 5.4

<u>Summative</u>: test #3

Time:	**Introduction:**
Part 1	Take up / check homework from lesson #9 and check for student's understanding. If needed, go over specific questions.

Time: **Part 2**	**Lesson:** <u>Chapter #5.2 Graphs of Reciprocal Trigonometric Functions</u> Given graphs of reciprocal trigonometric function, define all characteristics of cosecant, secant and cotangent and then relate back to the function of these graphs. Relate graphs of cosecant and secant to each other and let students recognize that they differ only from the placement of the graphs (e.g. secant is just $\pi/2$ “behind” cosecant), exactly like the relationship between sine and cosine. Also relating the graph of cotangent to its alternative form, $\frac{1}{\frac{\sin sin\, x}{\cos cos\, x}} \rightarrow \frac{\cos cos\, x}{\sin sin\, x}$, and recognize the reasoning behind cotangent’s asymptotes is because sine cannot equal to zero (opposite of tangent), and therefore for all points that sine does equal to zero, there is vertical asymptote. Also compare and contrast graphs of tangent versus cotangent. <u>Chapter #5.3 Sinusoidal Functions of the Form</u> $f(x) = a \sin sin\, / \cos cos\, [\, k(x - d)] + c$ Similar to all transformations obtained with other functions, define the significance behind the variables *a*, *k*, *d* and *c*. With this in mind, emphasis when the graph of sine is moved $\pi/2$ to the left, it equal to the graph of cosine (thus $\sin sin\, (x + \frac{\pi}{2}) = \cos cos\, x$). Relate this back to the co-function identities and how they relate. <u>Chapter #5.4 Solving Trigonometric Equations</u> Start by differentiating the different ways that students are able to solve trigonometric equations. Emphasize on using the quadratic method and solving by simplifying equations. Introduce *Symbolab* to solve for difficult equations.
Time: **Part 3**	**Wrap-up:** Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student’s learning needs for chapter #4 and chapter #5.

Notes, reminders & homework:

<u>Reminder</u>:

- Quiz #4 (to be written before / after current class)
- Test #3 (to be written before / after lesson #11)

<u>Homework</u>:
Chapter 5.2: # 4, 6 – 10
Chapter 5.3: # 1 – 8, 11, 14 - 17
Chapter 5.4: # 1 – 6, 10, 11

Lesson Reflection: